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Influence of molecular hydrogen on acetylene pyrolysis: Experiment and modeling

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ABSTRACT

The effect of molecular hydrogen on the formation of molecular carbonaceous species important for soot formation is studied through a combination of shock-tube experiments with high-repetition-rate time-of-flight mass spectrometry and detailed chemistry modeling. The experiment allows to simultaneously measure the concentration–time profiles for various species with a time resolution of 10 µs. Concentration histories of reactants and polyacetylene intermediates ($C_{2x}H_2$, $x = 1–4$) are measured during the pyrolysis of acetylene with and without H_2 added to the gas mixture for a wide range of conditions. In the 1760–2565 K temperature range, reasonable agreement between the experiment and the model predictions for C_2H_2, C_4H_2, C_6H_2, and C_8H_2 is achieved. H_2 addition leads to the depletion of important building blocks for particle formation, namely of polyacetylenes due to an enhanced consumption of important radicals by H_2, which are required for the fast build-up of carbonaceous material.

1. Introduction

Clarifying the details of soot formation under a wide range of conditions has been an active research topic for many years with the aim to develop models that can reliably predict soot formation, growth, and burnout. Acetylene is a major decomposition product of several hydrocarbons and plays an important role as a building block towards soot formation. Its thermal decomposition has been studied in a wide temperature range by several authors [1–3]. The most common approach for modeling soot formation relies on the HACA (hydrogen-abstraction, carbon-addition) mechanism [4]. Despite its prominent role, the influence of hydrogen on the formation of soot precursors and soot is not yet fully understood. Hydrogen can either be present as bonding partner in the fuel molecules, or as molecular hydrogen formed during pyrolysis or it can be present as part of the original gas mixture. Already in 1950, Arthur described that the reduction of the H-atom concentration in flames is accompanied with the suppression of flame luminosity, i.e., particle concentration [5]. Recent work aimed at elucidating the influence of hydrogen addition on soot formation and oxidation more systematically [6,7]. However, these theoretical studies so far lack experimental validation, and it is still controversial whether hydrogen promotes or inhibits soot formation [8–11].

In the last two decades many groups shed light on the impact of hydrogen on the soot build-up. Du et al. [12] observed a substantial decrease in the soot inception limit, i.e., a decrease in the soot particle inception rate with hydrogen addition in non-premixed counterflow ethylene, propane, and butane flames. The authors argued that the observed influence of H_2 could be due to chemistry but it could also be explained on purely physical grounds. Gülder et al. [13] demonstrated that for a non-premixed ethylene/air flame, in contrast to propane and butane, the addition of H_2 to the fuel suppresses the soot formation through both dilution and chemistry effects. The reduction of soot by hydrogen in propane and butane flames was suggested to be only due to dilution. In a recent study, Pandey et al. [14] investigated the influence of H_2 addition on soot formation and soot morphology in non-premixed laminar acetylene/air flames with various flow arrangements. They reported that the H_2 addition increases the centerline flame temperature by 50–100 K and that the soot volume fraction decreases with H_2 addition. These findings are in good agreement with results for ethylene, propane, and butane/air flames which have been explained based on the HACA mechanism.

Abian et al. [15] studied the effect of H_2 and typical components of recirculated exhaust gases (CO, H_2O, CO_2) on soot and other gaseous products, that are formed during the thermal decomposition of ethylene. The experiment was carried out in a flow reactor and the results were compared with simulations based on a

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gas-phase kinetics model. They found that the addition of H_2 chemically alters the reaction pathways leading to the formation of soot, and consequently suppresses soot production, due to its direct participation in the HACA mechanism [10]. The effect of H_2 on the oxidation of methane was studied experimentally and modeled with a kinetics model by Cong et al. [16]. They found that H_2 accelerates methane oxidation by producing O, H, and OH. Ruiz et al. [17–19] have studied the soot yield from C_2H_2 and C_2H_4 pyrolysis and its properties in a flow reactor as a function of temperature, initial concentrations, and residence time. More recent investigations dealing with the effect of hydrogen on soot formation can be found in [20–26]. Eremin et al. published a detailed experimental study to elucidate the role of hydrogen on particle growth behind shock waves where temperature, particle sizes and induction times were measured [27]. It was found that the measured temperatures were similar for both cases (with and without H_2 addition) and that H_2 addition increases the induction times and decreases both the particle sizes and optical density of the soot-containing mixture. Böhm et al. [28] investigated the effect of hydrocarbon-bonded (C_2H_2) and molecular hydrogen (H_2) on precursor and particle formation in the pyrolysis of C_3O_2 in argon behind reflected shock waves. It was found that H_2 reduces the particle volume fraction, the mean particle diameter, the particle number density, and the maximum temperature rise of the mixture.

All these works have in common: They investigate the effect of molecular hydrogen in flames or in flow reactors where the main targets were soot particles in the late stage of formation. Shock tubes were also used to elucidate the effect of H_2 on acetylene, but the focus was also set on the consequence of adding H_2 on the optical density of the reaction mixture, on particle sizes, as well as gas temperature. In the present work, the influence of H_2 on carbonaceous reaction intermediates that are important for soot precursor formation in acetylene pyrolysis has been studied in a shock tube that prevents transport phenomena that may interfere with chemical processes in flames and flow reactors. The study contributes to the understanding of soot formation in its early stage by time-resolved measuring absolute concentrations of polyacetylenes considered as potential precursors leading to soot formation. Additionally, a detailed kinetics model was developed to explain the experimental observations.

Shock tubes that are equipped with a variety of modern diagnostics are versatile tools that are widely used for gas-phase reaction kinetics studies [29,30] enabling the investigation of high-temperature reactions on short time scales (up to a few milliseconds). Fast detection methods are required, that provide time-resolved information about the change of composition of the investigated mixture within the test time.

A high-repetition-rate (HRR) time-of-flight mass spectrometer (TOF-MS) can simultaneously measure the concentration–time profiles for numerous species with approximately one measurement every 10 μs. Combinations of shock tubes with TOF-MS are established for the study of complex reaction systems [2,31,32], in particular those that involve soot formation. In the present study, we used a shock tube coupled to a HRR-TOF-MS to investigate the thermal decomposition of acetylene with and without hydrogen behind reflected shock wave in details.

Polyacetylenes are assumed to be "pre-soot" species in pyrolysis and oxidation of most fuels. The direct and simultaneous detection of them gives the possibility to investigate the influence of hydrogen towards soot formation in its earlier stages.

2. Experiment

The experiments are conducted in a stainless-steel diaphragm-type shock tube. Both the driver and driven section have an inner diameter of 80 mm. The driver section has a length of 2.5 m and the driven section measures 6.3 m. Aluminum sheets with a thickness of 50–100 μm are used as diaphragms between the driver and the driven section. Both driver and driven sections are simultaneously pumped down to 4×10^{-2} mbar (Edwards Dry Star, QDP 80). Via a by-pass line, this pump is also used to evacuate the 50 l vessel that is used for preparing the investigated gas mixtures. The driver section is filled via a magnetic valve with the driver gas (here: helium). The pressure in the driver section is measured with a pressure gauge (Keller, Mano 2000) which is also used to measure the diaphragm rupture pressure. The inlet pressure of the reactive mixture (p_1) in the driven section is measured with a pressure sensor (Edwards, Trans 600 AB) usable for pressures up to 1000 mbar. A set of four pressure transducers (PCB-112A05) are positioned at equal distances of 150 mm on the top of the driven section of the shock tube. The fourth transducer is located 150 mm from the end plate. A fifth pressure transducer located close to the end plate measures the reflected-shock pressure. The signals of these pressure transducers are amplified with charge amplifiers (Kistler, Kiag Swiss 5001) and detected by an oscilloscope to determine the shock wave velocity. The pressure and temperature behind the reflected shock wave (p_5 and T_5) are calculated from shock wave equations [33] using the initial pressure and temperature in the driven section (p_1 and T_1) and the velocity of the incident shock wave which is determined from the pressure transducer signals. Attenuation values of the shock velocities give uncertainties in the determined post-shock temperature of around ±15 K.

A precision-manufactured conical nozzle (Frey, 45 μm diameter) is located in the center of the end flange that separates the driven section and the TOF-MS. The distance between the nozzle and the ionization section of the TOF-MS is 8 mm. The TOF-MS (Kaesdorf) can operate at high repetition rates (up to 150 kHz). Both, short flight distance in the TOF-MS and high acceleration voltage ensure short flight times which avoids overlapping of signals from consecutive cycles up to masses of 240 u at 100 kHz. The TOF-MS is equipped with an electron impact ion source with energies in the 5–85 eV range. The maximum kinetic energy of the ions is 10 keV, which is high enough to generate signals for molecular masses up to 1000 u (then, with a maximum repetition rate of 50 kHz). The apparatus is used in reflectron mode and the ions are detected with a micro-channel plate (MCP) detector. More details about the experiment are given in [34].

When the shock wave passes the first pressure transducer (the one farthest from the end flange), a TTL signal triggers the ionization in the TOF-MS and the data of the subsequent mass spectra are acquired by an oscilloscope (Agilent Acquiris). Several full mass spectra are thus detected already before the arrival of the shock wave at the sampling nozzle, which then helps to define the exact zero time of the shock arrival from the sudden rise in signal. Figure 1 shows raw data obtained by the TOF-MS. The sudden increase of the signal intensity is due to the pressure increase behind the reflected shock wave which is accompanied by an increase of the mass flow into the mass spectrometer. Because of the restricted time resolution in this figure, each full mass spectrum is visible as a single peak only with the maximum signal due to the bath gas neon. Neon is chosen as bath gas because of its comparably small ionization cross-section to prevent overloading of the MCP detectors; but because of its high concentration (>90%) it still provides the strongest signal.

The mass-to-charge ratio m/z of the detected ions is determined for each peak from its time of occurrence t.

$$m/z = a(t - b)^2 \tag{1}$$

The constants a and b are determined from calibration measurements with noble gas mixtures. Figure 2 shows a room-temperature

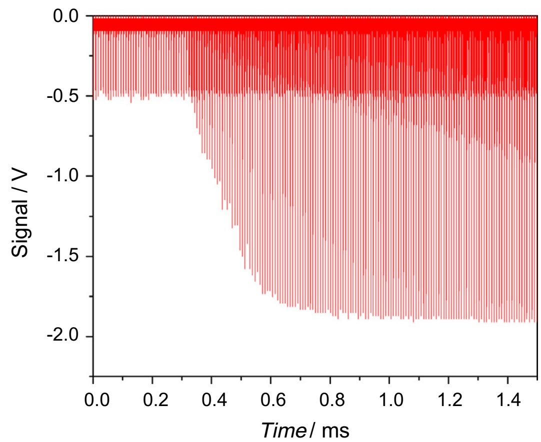

Fig. 1. Raw HRR-TOF-MS data. The repetition rate is 100 kHz. The mixture is 5% C_2H_2 and 1% Ar in Ne, p_5 = 0.95 bar and T_5 = 2225 K.

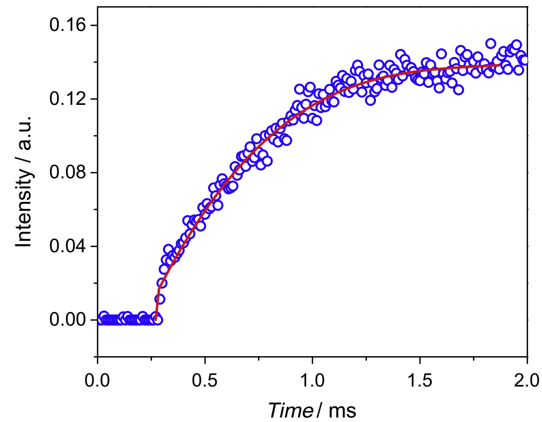

Fig. 3. Temporal variation of argon peak intensities for subsequent mass spectra for an experiment with 5% C_2H_2 and 1% Ar in Ne at p_5 = 1.08 bar and T_5 = 1995 K. The polynomial fit (solid line) is used for sensitivity-correction of all other species signals.

spectrum measured with 100 kHz ionization repetition rate and averaged over 400 individual spectra.

The expansion of the gas suffers from gas-dynamic effects in the initial phase after the shock arrival which leads to a reduced sensitivity in this initial phase (visible from the steady signal increase in the first ~30 spectra in Fig. 1). Additionally, at longer test times the sensitivity decreases due to ion losses in the mass spectrometer because of slowly rising pressure in the vacuum chamber. Therefore, Ar (or Kr in case of eventual overlapping signals on the mass of Ar) is added to the gas mixture in the shock tube at low concentrations (1%) and its signal intensity is used as a reference. Figure 3 shows the time-dependent signal of the internal standard (Ar) for an experiment with T_5 = 1995 K and p_5 = 1.08 bar. All the species signal profiles are then divided by the Ar signal intensity profile to correct for the time-varying sensitivity of the instrument. Before the correction, the Ar signal is fitted with a polynomial function to avoid deterioration of the signal-to-noise ratio of the measured species signals by the scatter of the Ar signal.

The gas mixtures are prepared in a mixing vessel (50 l volume). In this study the measurements are performed with two different concentrations of C_2H_2, 2% and 5% in Ne. For the 5% case, the nozzle was frequently blocked in the experiments as a result of soot formation apparently due to surface reactions on the nozzle. Typically, a trade-off between nozzle diameter and concentration of the precursor is chosen to avoid this issue. Therefore, most experiments are carried out with 2% C_2H_2 but some measurements with 5% are also included for those experiments where the nozzle was not blocked. For the measurements with H_2, mixtures of 2% and 5% C_2H_2 with 2% and 4% H_2 in Ne are prepared. All mixtures additionally contain 1% Ar as internal inert standard. After preparing the mixture, it is allowed to homogenize for several hours in the mixing vessel, typically over night. The experimental conditions are given in Table 1.

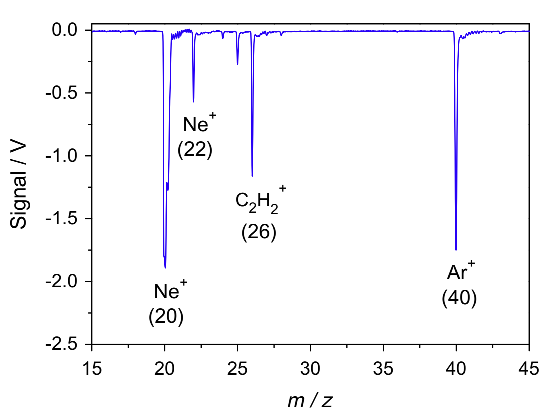

Fig. 2. Average mass spectrum of C_2H_2 at room temperature. The repetition rate of ionization was 100 kHz and the energy of the ionizing electrons was 45 eV. Mixture: 1% C_2H_2, 2% Ar in Ne.

3. Modeling

The acetylene reaction mechanism employed here has been validated in several publications before [35–38]. The C_1 chemistry relies on the work of Sojka [39], the C_2 reactions are assembled from Heghes [40], while the core of the reactions of the C_3 to C_8 species follows the work of Böhm et al. [41]. The polyaromatic hydrocarbon chemistry as given by Frenklach et al. [4] for the HACA route, as well as PAH channels as proposed and validated by Böhm and Jander [42] and by Naydenova [43] for the incorporation of combinative steps of aryls were employed. Besides, the C_2 channel for the growth of aromatic hydrocarbons as proposed by Gu et al. [44] with the rate coefficient from Becker et al. [45] for the mentioned reaction type are taken into consideration. The model also contains carbon-cluster formation for particles up to C_{84} as published in [22], with improvements as presented and validated in [28]. The calculations were performed under constant volume assumption using the CHEMKIN package [46]. The model describing polyacetylenes chemistry until $C_{12}H_2$ is presented in the appendix. The full model is available from the authors.

4. Results and discussion

The pyrolysis of acetylene (C_2H_2) is studied at temperatures between 1760 and 2565 K and pressures between 0.79 and 1.28 bar. Figure 4 shows a post-shock mass spectrum for an experiment with T_5 = 1995 K and p_5 = 1.08 bar. It exhibits the typical fragmentation pattern of C_2H_2 with electron-impact ionization

Table 1
Experimental conditions.

Composition	Pressure (bar)	Temperature (K)
2% C_2H_2 + 1% Ar + Ne	0.9–1.3	1855–2310
2% C_2H_2 + 2% H_2 + 1% Ar + Ne	0.9–1.5	1895–2410
2% C_2H_2 + 4% H_2 + 1% Ar + Ne	0.95–1.1	1995–2430
5% C_2H_2 + 1% Ar + Ne	0.95–1.1	1995–2430
5% C_2H_2 + 2% H_2 + 1% Ar + Ne	0.75–1.3	1950–2250
5% C_2H_2 + 4% H_2 + 1% Ar + Ne	0.75–1.2	1750–2300

(here: 45 eV, cf. Fig. 2). The additional peaks visible at masses higher than $m/z = 40$ are associated with products of the thermal decomposition of C_2H_2. The post-shock spectrum shows signals that are attributed to higher polyacetylenes (C_4H_2, C_6H_2, and C_8H_2). The first aromatic structure (e.g., benzene), however, was not detected. The concentration of benzene is below the threshold of the spectrometer sensitivity (1.7×10^{-4}). This is in agreement with the calculations that predict very small mole fractions of the aromatic species ($x_{aromatic} < 10^{-8}$). Only in the experiments with 5% C_2H_2, traces of C_4H_3, which is an important intermediate to form the first aromatic ring (forming phenyl via C_2H_2 addition) is found.

For quantitative analysis of the signal intensities the TOF-MS must generally be calibrated for each species individually. This is doable only for stable species or for species where a reliable quantitative formation reaction is available. For other species, semi-quantitative data is available that provide reliable information about the temporal variation, but not absolute concentrations. To calibrate C_2H_2 in shock-heated measurements with known initial concentrations, studies are performed at relatively low temperatures where C_2H_2 does not decompose and signal intensities are converted to absolute concentrations. In case of polyacetylenes, however, diacetylene is very unstable. Therefore, the calibration factors m ($I_i = mc_i$), for C_4H_2 and C_6H_2 were determined by taking into account the calibration factor ratio of C_2H_2:C_4H_2:C_6H_2 (1:2:3), where I_i and c_i are the intensity and the corresponding concentration of the species i. These factors are in line with the respective measured and calculated ratios of ionization cross sections from the NIST database [47] but are slightly different from the ratios adopted in [2]. The calibration factors for the stable species were determined from measurements with 3–5 different concentrations of the respective species in the calibration mixtures. The interpolation yields calibration factors with uncertainties of around 10% that are then used to derive concentrations from the signal intensities. The experimental data are compared with results of simulations based on two previous models [2,3] as well as the model used in the present work. Figure 5 shows an example of concentration–time profiles for several species in a reacting mixture of 2% C_2H_2, 1% Ar, and 97% Ne at $T_5 = 2240$ K and $p_5 = 0.96$ bar. The concentration–time profiles show that acetylene is not completely converted within the test time. The signal intensities of the polyacetylenes decrease with increasing molecular weight ($C_4H_2 > C_6H_2 > C_8H_2$) and reach a plateau at longer times. The examination of the calibrated profiles with respect to the carbon mass indicates well-balanced species concentrations within the experimental uncertainties. Moreover, the maximum C_6H_2 concentration is 20 times smaller than the initial C_2H_2 concentration which is in line with previous observations [2]. The measured concentration–time profiles are compared with simulations based on the three kinetics

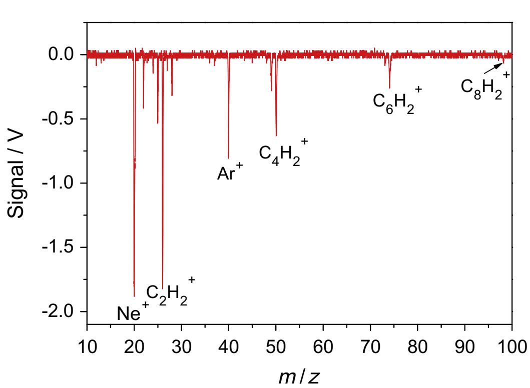

Fig. 4. Post-shock ($p_5 = 1.08$ bar, $T_5 = 1995$ K) mass spectrum for 5% C_2H_2 and 1% Ar in Ne.

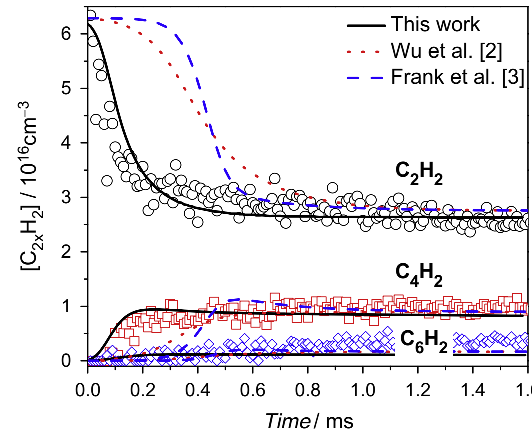

Fig. 5. Measured (symbols) and simulated (lines) concentration–time profiles of the main species during C_2H_2 pyrolysis (2% C_2H_2, 1% Ar, 97% Ne, $T_5 = 2240$ K, $p_5 = 0.96$ bar). Simulations based on the models from [2,3] are compared with the model of this work.

models in Fig. 5. The Frank and Just [3] and Wu et al. [2] models predict a longer induction time for acetylene depletion and a later appearance of the products compared to the model used in the present work. The latter shows significantly better agreement with the experiment. This trend is observed throughout the whole range of temperatures and reactant concentrations. Simulations based on all three models, however, coincide with the experiment at longer reactions times.

The influence of H_2 on the C_2H_2 decomposition chemistry is studied in mixtures containing 2% C_2H_2 with 2% and 4% H_2. Preliminary tests showed that the selected initial concentrations were the best choice to achieve an acceptable signal-to-noise ratio and to avoid blockage of the nozzle. Using higher concentrations than 4% H_2 was not straightforward, because in this case, the consumption of C_2H_2 is strongly decreased leading to very low concentrations of polyacetylenes, that could only be measured with significant error. Figure 6 shows concentration–time profiles for mixtures without and with molecular hydrogen ($x(H_2) = 0\%$, 2%, and 4%) at approximately the same experimental conditions ($T_5 \approx 2300$ K, $p_5 \approx 1$ bar). In the presence of H_2, less C_2H_2 (almost 50% less than in the case where H_2 is not added) is consumed and half of the diacetylene is produced only. C_6H_2 and C_8H_2 are not detected when H_2 is added within the detection limit (i.e., the concentrations are reduced by more than a factor of 50 compared to the cases without H_2). The concentration of C_4H_2 with addition of 2% H_2 is higher than that for the case without H_2 addition because of the difference of post-shock pressures.

The corresponding simulations for C_2H_2 and C_4H_2 for the same conditions with and without H_2 are also shown in Fig. 6. Although the consumption of C_2H_2 is underpredicted by the model, calculations and measurements showed the same trend for the influence of H_2 on the species concentration variation, namely reduced consumption of C_2H_2 and reduced formation of polyacetylenes, here, exemplarily demonstrated for C_4H_2. Again, a similar trend was observed for the full temperature range and concentration variations investigated in this work. Figure 7 show three additional experiments with higher acetylene concentration (5%) and with variable H_2 concentration (0–4%). Again all profiles are well reproduced by the model that shows that H_2 addition inhibits the formation of higher acetylenes.

In exploring the reasons for this behavior, a reaction path analysis is carried out. During the pyrolysis of acetylene at high temperatures the following two reactions compete ((R1) and (R2)):

$$C_2H_2 + M = C_2H + H + M \quad \text{(R1)}$$

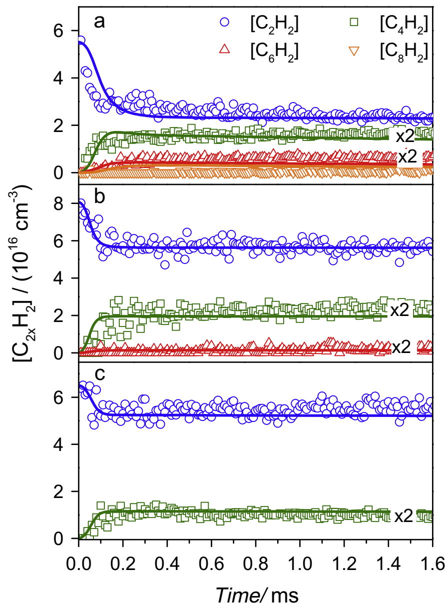

Fig. 6. Comparison of the measured and calculated concentration–time profiles of $C_{2x}H_2$ species for a mixture of 2% C_2H_2, 1% Ar, and 97% Ne at T_5 = 2310 K and p_5 = 0.88 bar (a), a mixture of 2% C_2H_2, 2% H_2, 1% Ar, and 95% Ne at T_5 = 2315 K and p_5 = 1.29 bar (b) and a mixture of 2% C_2H_2, 4% H_2, 1% Ar, and 93% Ne at T_5 = 2315 K and p_5 = 1.04 bar (c). Solid lines are results of the simulations from this work (see text).

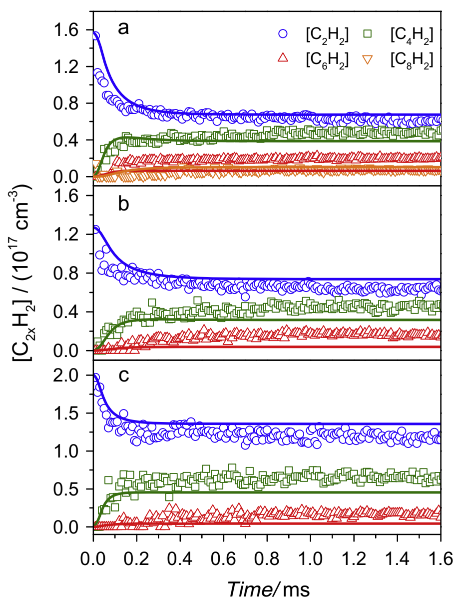

Fig. 7. Comparison of the measured and calculated concentration–time profiles of $C_{2x}H_2$ species for a mixture of 5% C_2H_2, 1% Ar, and 97% Ne at T_5 = 2225 K and p_5 = 0.95 bar (a), a mixture of 5% C_2H_2, 2% H_2, 1% Ar, and 95% Ne at T_5 = 2195 K and p_5 = 0.76 bar (b) and a mixture of 5% C_2H_2, 4% H_2, 1% Ar, and 93% Ne at T_5 = 2215 K and p_5 = 1.21 bar (c). For visualization purposes, all concentrations of higher polyacetylenes were multiplied by factor 2. Solid lines are results of the simulations from this work (see text).

$$C_2H_2 + C_2H_2 = C_4H_3 + H \tag{R2}$$

Further H-elimination reaction of C_4H_3 produces C_4H_2 (R3). Additionally, C_4H_2 is produced by addition of the acetyl radical (C_2H) to C_2H_2.

$$C_2H_2 + C_2H = C_4H_2 + H \tag{R4}$$

Further C_2H addition to C_4H_2 and its reaction products via reaction type (R4) forms the next higher polyacetylenes C_6H_2, C_8H_2, etc. In the experiments, C_4H_3 can be distinguished from C_4H_2 but is observable in experiments with initial concentrations of 5% C_2H_2 only. In this case, the intensity is about a factor 10 smaller than that of the major product C_4H_2 but higher than the peak intensity on the same mass caused by the isotopic signature of ^{13}C with an abundance of 4.5% for C_4H_2. Therefore, the identified mass at 51 clearly reflects the C_4H_3 species.

A reaction-path analysis was performed to determine the fate of acetylene and the reaction channels that are important for the formation of higher polyacetylenes. The results are exemplarily shown in Fig. 8 for a reaction time of 0.1 ms and for a mixture of 5% C_2H_2 at 2225 K and 0.95 bar. It can be clearly seen that acetylene is mainly consumed by reaction (R4) and

$$C_2H_2 + H = C_2H + H_2 \tag{R5}$$

Further formation of polyacetylenes ((R6)–(R8)) exhibits increasingly lower importance regarding C_2H_2 consumption:

$$C_2H_2 + C_4H = C_6H_2 + H \tag{R6}$$

$$C_2H_2 + C_6H = C_8H_2 + H \tag{R7}$$

$$C_2H_2 + C_8H = C_{10}H_2 + H \tag{R8}$$

In addition, simulations were performed using recent models of Chernov et al. [48] and Ranzi [49] to show their performance towards the prediction of polyacetylenes. The results are shown in Fig. 9 for an example of 5% C_2H_2 balanced with neon and at the same post-shock condition. The model of Chernov et al. predicts a slower reaction progress compared to our model and produces more C_4H_2 and less C_6H_2. Also, the plateau seen in the experimental data was not well reproduced. It should be noted that the model was developed to predict the gas-phase growth of PAH in methane, ethylene, and ethane flames but contains a submechanism for acetylene.

The model of Ranzi, however, reaches the equilibrium plateau fast, but the predicted concentrations deviate from the measured ones for C_2H_2 and C_4H_2. We identified the most important reactions of the three models for acetylene consumption to analyze the difference in their predictions. In the Chernov mechanism, the depletion of C_2H_2 is mainly controlled by reaction (R1) C_2H_2 + M = C_2H + H + M, (R5) C_2H_2 + H = C_2H + H_2 and (R9) C_2H_2 + C_2H_2 = C_4H_2 + H_2. But these reactions cannot explain the discrepancies of the three models. It is rather the availability of the H-atom pool in each model that promotes further chain reactions.

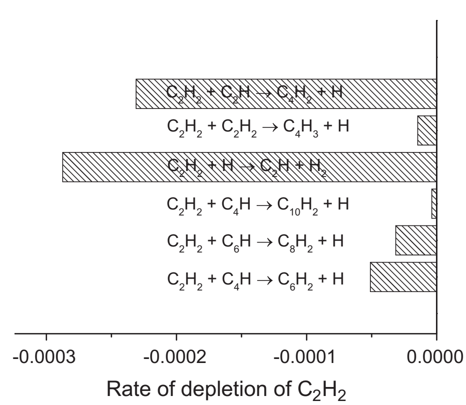

Fig. 8. Rate of depletion of acetylene at 2225 K and 0.95 bar and a mixture of 5% C_2H_2, 1% Ar, and 97% Ne.

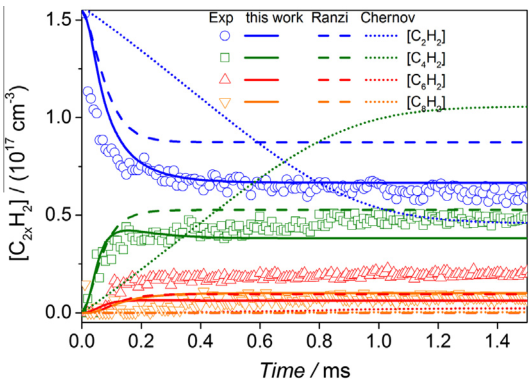

Fig. 9. Simulations based on three models (this work, Chernov et al. [48], and Ranzi [49]) for an experiment with 5% C_2H_2 and 1%Ar in Ne (T_5 = 2225 K, p_5 = 0.95 bar).

The main H-atom sources identified in our model are:

$$C_2H_2 + C_2H_2 = C_4H_3 + H \quad \text{(R2)}$$

$$C_4H_3 = C_4H_2 + H \quad \text{(R3)}$$

with rate coefficients of $2.00 \times 10^{13} \exp(-44500\ \text{cal}/RT)$ and $1.70 \times 10^{18}(T/\text{K})^{-1.23} \exp(-42538\ \text{cal}/RT)$, respectively. The Chernov model contains only the much slower reaction $2C_2H_2 = H_2CCCCH + H$ with a rate coefficient of $2.00 \times 10^9 \exp(-29107\ \text{cal}/RT)$.

Figure 10 shows the H-atom concentrations predicted by the three models. Our model produces within 50 µs sufficient H atoms to actively initiate further chain reactions whereas in the model of Chernov the H-atom concentration increases steadily. In contrast, in the Ranzi model, the initial H-atom concentration is lower than in our mechanism but the final values are close to each other. The H-atom initiation reaction (R1) in the Ranzi model has a rate coefficient of $k = 2.00 \times 10^{16} \exp(-81500\ \text{cal}/RT)$ which is at 2225 K a factor of five lower than the value in our mechanism. When this rate coefficient is replaced with our value, both H-atom concentration profiles coincide (dashed line in Fig. 10).

Although the reactions (R5) $C_2H_2 + H = C_2H + H_2$ and (R10) $C_2H_2 + C_2H = C_4H + H$ exhibit high rates of production and removal of H atoms, their effect on the H-atom concentration does not influence the reaction progress because their contributions are antagonistic and cancel. Hence, only reaction (R2) and (R3) play a major role.

When incorporating both reactions ((R2) and (R3)) with the respective reaction coefficients in the Chernov model together with the rate coefficient for reaction (R5) $2.0 \times 10^9 (T/\text{K})^{1.64} \exp(-30289\ \text{cal}/RT)$ both models agree in the C_2H_2 decay although the (final) equilibrium concentrations still differ, apparently because of differences in thermodynamics data.

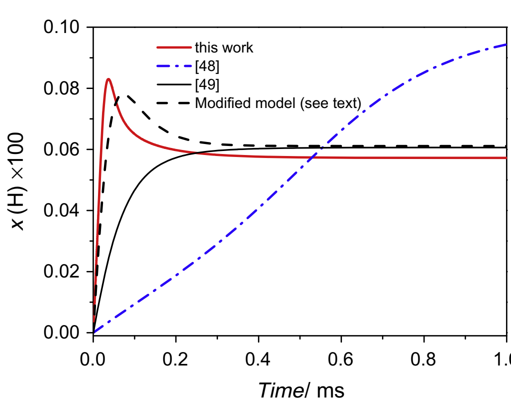

Fig. 10. Simulated H-atom concentration for a mixture of 5% C_2H_2 balanced in neon. T_5 = 2225 K and p_5 = 0.95 bar.

As shown above, in the experiments and simulations with added H_2, decreased C_2H_2 consumption and decreased concentrations of larger polyacetylenes are found. As deduced from the modeling, the addition of H_2 consumes C_2H by an increased rate of the reverse reaction (R−5).

$$C_2H_2 + H = C_2H + H_2 \quad \text{(R5)}$$

Therefore, less C_2H is available to form C_4H_2 via (R4) in these mixtures. Furthermore, hydrogen atoms whose concentration also increases when H_2 is added enhance the rate of dissociation of reaction type (R−4) for C_4H_2 and higher polyacetylenes leading to a decreased concentration of the latter compared to the pure C_2H_2/Ne pyrolysis. The inhibiting effect of H_2 is confirmed in the experiment for acetylene and the polyacetylenes. These results are in agreement with those of [27] where a decrease of particle volume fraction and particle sizes was observed.

The addition of H_2 causes also the formation of higher concentrations of aliphatic hydrocarbons by the reaction of C_2H_2 with H_2:

$$C_2H_2 + H_2 = C_2H_3 + H \quad \text{(R11)}$$

$$C_2H_2 + H_2(+M) = C_2H_4(+M) \quad \text{(R12)}$$

This trend is also observed in the experiments where the formation of hydrocarbons such as C_2H_4 is enhanced with hydrogen addition.

The concentrations of the aromatic compounds are below the detection limits of our experiment. This is in agreement with the simulations where the mole fraction of benzene is below 10^{-8}, the mole fractions of larger aromatic species are even smaller. Thus, regarding the influence of H_2 on the formation of polyaromatic hydrocarbons (PAH) in C_2H_2 pyrolysis, only modeling studies can be carried out. The simulations predict a decreasing concentration of these species by the addition of H_2. This is due to the important role of aromatic radicals (PAH$^\cdot$) for the build-up of large PAH structures. The concentration of these radicals is influenced by H_2 via reactions of the type

$$PAH + H = PAH^\cdot + H_2 \quad \text{(R13)}$$

The availability of larger H_2 concentrations favored the reverse reaction (R −13) and thus decreased the concentrations of the PAH$^\cdot$ radicals which consequently results in a less efficient way for PAH formation by C_2H_2 addition through reactions of the type

$$PAH^\cdot + C_2H_2 = PAHC_2H + H \quad \text{(R14)}$$

Because of the abundance of H_2, more aromatic radicals are removed via (R −13) before adding C_2H_2 via (R14).

In comparison with the pure pyrolysis case, where H_2 and H atoms can also be also formed from hydrocarbon-bonded hydrogen, the addition of molecular hydrogen leads to a faster availability of reactive species (H, radicals) which cause a depletion of building blocks of particle formation, such as polyacetylenes, aromatic species, and carbon clusters. The equilibrium (final) value of C_2H_2 seems to be reached faster when H_2 is added.

5. Summary and conclusions

Thermal decomposition of C_2H_2 was studied behind reflected shock waves in the 1760–2565 K temperature range for pressures between 0.75 and 1.23 bar. Concentration–time profiles for C_2H_2 and the main intermediates ($C_{2x}H_2$, x = 2–4) were measured by high-repetition-rate time-of-flight mass spectrometry (HRR-TOF-MS). Benzene and higher polyacetylenes (x > 4) were not detected within the detection limits of the experiment. The experimental results were compared to simulations based on a recently presented model and two models from the literature [2,3]. While the prediction was good for most detected species after longer test times with all three models, the new model provided significantly improved simulations for the concentration–time profiles within the first 500 μs. The results were also compared with two additional models with large C_2H_2 submechanisms from the literature [48,49]. Reaction rate analysis has shown that the reactions of H atom have a significant impact on the kinetics of acetylene and polyacetylenes. The very good agreement of the experiments and the calculations show that these reactions are obviously very well described in our mechanism.

The impact of H_2 on carbonaceous reaction products, important for soot precursor formation was as followed: The presence of H_2 reduced the consumption of acetylene, and reduced the concentration of the respective polyacetylenes and of PAHs. This is attributed to an enhanced consumption of the crucial carbon-adding (di)radicals C_2H, PAH$^{\cdot}$, C_2, especially required for the fast build-up of carbonaceous material of particle precursors, resulting in species of less reactivity.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.combustflame.2014.03.012.

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